Multi-Class Text Classification: A Comparison of Word Representations and ML/NN Models

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***Abstract*—Multi-class text classification aims to automatically assign each text instance to one category among multiple possible classes, making it a central task in many NLP applications.This study examines how different word representation techniques and classification models influence performance on a ten-class question–answer dataset. Exploratory data analysis is first conducted to understand class distribution and text length characteristics, which informs the design of preprocessing steps. Two word representation methods are evaluated: TF–IDF and Skip-gram embeddings. TF–IDF is paired with a traditional machine learning classifier and a deep neural network, while Skip-gram embeddings are used with multiple neural architectures, including recurrent and bidirectional recurrent models.All models are manually tuned using validation performance and evaluated on a held-out test set using standard multi-class metrics. The results show that neural models using Skip-gram embeddings consistently outperform TF–IDF-based approaches, with bidirectional recurrent architectures providing the strongest overall performance. In contrast, simpler recurrent models struggle to capture the structure of longer question–answer texts. These findings highlight the importance of contextual sequence modeling for multi-class text classification tasks.**

***Index Terms*—Multi-class Text Classification, Natural Language Processing, TF-IDF, Skip-gram, Neural Networks, Recurrent Neural Networks, Bidirectional LSTM, Macro F1-score**

I. Introduction

Automatic categorization of text into predefined classes is a core problem in Natural Language Processing and underpins many real-world language understanding systems. Such systems are commonly used in applications including question answering, information retrieval, topic classification, and content organization. The effectiveness of these applications depends on the ability of classification models to accurately capture semantic meaning and contextual relationships within text. As textual data continues to grow in volume and complexity, designing models that can reliably distinguish between multiple categories has become increasingly important.

The choice of word representation and model architecture plays a critical role in determining classification performance. Traditional approaches often rely on frequency-based features, while neural approaches leverage distributed representations and sequence modeling to better capture contextual information. The purpose of the study is to examine how these different design choices affect performance on a multi-class question–answer classification task.

Therefore ,the goal of this project is to compare classical and neural models under two representation paradigms: TF–IDF and Skip-gram embeddings. The comparison focuses on the required model set and evaluation metrics, and the final analysis highlights the best- and worst-performing systems and includes a direct comparison between the best ML approach and the best NN approach

II. Dataset and Exploratory Data Analysis (EDA)

A. Dataset Overview

The dataset contains QA-style text samples with labels across 10 classes (Business & Finance, Computers & Internet, Education & Reference, Entertainment & Music, Family & Relationships, Health, Politics & Government, Science & Mathematics, Society & Culture, Sports).The dataset is provided in separate training and test CSV files, and the test split is reserved strictly for final evaluation in accordance with the project rules. [**?**]

B. Class Distribution

The class distribution plot indicates that the dataset is broadly balanced across categories. This makes accuracy a meaningful metric, however,t Macro F1-score is also significant because it evaluates performance uniformly across all classes.

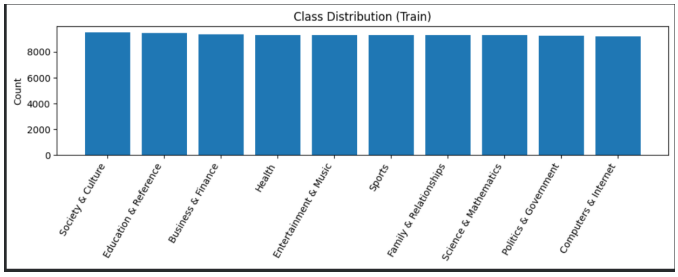


Fig. 1. Class Distribution(Train)

C. Text Length Distribution

Word count analysis shows a right-skewed, long-tailed distribution. Most samples fall within a moderate length range, while a smaller portion contains much longer text. This observation motivates the use of a fixed maximum sequence

length for neural models so that training remains efficient and inputs are consistent.

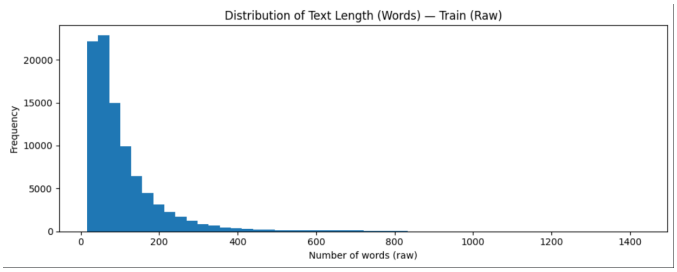


Fig. 2. Train split: Word Count Distribution

The dataset is broadly balanced and contains realistic variation in text length, making it suitable for comparing ML and NN models under consistent experimental settings.

III. Methodology

This section describes the overall experimental framework adopted in this project, including exploratory data analysis, text preprocessing, word representation techniques, model architectures, and evaluation strategy. All experiments were conducted using a consistent training–validation–testing pipeline to ensure fair comparison across different models and representations.

A. Text pre-processing

Based on the findings from EDA, a standardized preprocessing pipeline was applied to all textual data. Each question was first cleaned by removing HTML tags, punctuation artifacts, and extraneous symbols present in the raw dataset. The text was then converted to lowercase to reduce vocabulary sparsity and ensure consistency across samples.

Stopword removal and stemming or lemmatization were intentionally not applied. Since the dataset consists of questions, interrogative words and functional terms often carry important semantic information for classification. Removing such tokens could negatively impact performance. The same preprocessing pipeline was applied consistently to the training, validation, and test sets to maintain experimental fairness.

B. Word Representation Techniques

Two types of word representation methods were used to evaluate the impact of lexical and semantic features on classification performance.

- **TF–IDF (for ML and DNN):** TF–IDF is used to construct sparse lexical feature vectors. These vectors are well-suited for linear models and feedforward neural models operating on high-dimensional sparse inputs.
- **Skip-gram embeddings (for all NN models):** Skip-gram embeddings are trained locally and used to initialize embedding inputs for neural models. These embeddings provide dense semantic representations intended to support sequential modeling.

C. Model Architectures (Required Set)

To study the effect of different representations and learning models, a total of nine experiments are conducted using TF–IDF and Skip-gram word representations.

- **TF–IDF–Based Models:**
  - **TF–IDF + Logistic Regression:** This model uses TF–IDF features as input to a Logistic Regression classifier, serving as the required traditional machine learning baseline. It learns linear decision boundaries based on weighted word frequencies.
  - **TF–IDF + Deep Neural Network (DNN):** In this approach, TF–IDF vectors are fed into a feedforward deep neural network, allowing the model to learn non-linear patterns from the sparse feature representation.
- **Skip-gram–Based Neural Models:**
  - **Skip-gram + Deep Neural Network (Pooled):** This model uses Skip-gram word embeddings, which are aggregated into a fixed-length representation through pooling and passed to a deep neural network for classification.
  - **Skip-gram + SimpleRNN:** This model applies a Simple Recurrent Neural Network to sequences of Skip-gram embeddings, capturing basic sequential information without gating mechanisms
  - **Skip-gram + GRU:** In this architecture, Skip-gram embeddings are processed using a Gated Recurrent Unit, which improves sequence modeling by controlling information flow through gates.
  - **Skip-gram + LSTM:** This model uses Long Short-Term Memory networks with Skip-gram embeddings to capture long-term dependencies in text sequences.
  - **Skip-gram + Bidirectional SimpleRNN:** This variant processes Skip-gram embeddings in both forward and backward directions using a bidirectional SimpleRNN, enabling the use of full sequence context.
  - **Skip-gram + Bidirectional GRU:** This model combines Skip-gram embeddings with bidirectional GRU layers, allowing contextual information from both past and future tokens to be considered.
  - **Skip-gram + Bidirectional LSTM:** This architecture applies bidirectional LSTM layers to Skip-gram embeddings, providing the most expressive sequence modeling by capturing long-range dependencies in both directions.

D. Training Strategy and Hyperparameter Tuning

All models are trained using a consistent experimental setup, and hyperparameters are manually tuned in accordance with the project requirements. Model selection is guided by validation performance, ensuring that tuning decisions are based on generalization ability rather than test-set results. For each model, a subset of key hyperparameters—such as regularization strength for classical models and architectural or optimization parameters for neural networks—is adjusted during

experimentation For TF–IDF–based models, tuning focuses on selecting appropriate regularization and network capacity to balance model complexity and performance. For Skip-gram–based neural models, training configurations are chosen to effectively handle sequence length, hidden representation size, and regularization through dropout. Training stability and convergence behavior are monitored, and early stopping is applied to prevent overfitting and unnecessary training. All neural models are optimized using standard gradient-based optimization techniques as implemented in the notebook, and the best-performing configurations on the validation set are selected for final evaluation on the test set.

E. Evaluation Metrics

Model performance is evaluated using a set of standard metrics suitable for multi-class text classification. Accuracy is used to measure overall prediction correctness across all classes. Macro-averaged F1-score is reported to assess balanced performance by giving equal importance to each class, regardless of class frequency.

To provide a more detailed view of model behavior, confusion matrices are generated to analyze class-wise prediction patterns and identify commonly confused categories. In addition, classification reports are used to summarize precision, recall, and F1-score for each class, offering further insight into strengths and weaknesses at the class level. These metrics together ensure a comprehensive and fair evaluation of all models.

F. Result

This section presents a comparative analysis of all implemented models using the required evaluation metrics. The results highlight performance differences across word representations and model architectures, with particular emphasis on identifying the best- and worst-performing approaches.

G. Quantitative Performance Comparison

Table I summarizes the final test performance (Accuracy and Macro F1) for all nine required experiments. In the notebook output, the models are already sorted by Macro F1, which supports clear comparison.

H. Key Observations

The following observations summarize the main trends observed in the experimental results:

- The strongest overall performance is achieved by SkipGram+BiLSTM, indicating that bidirectional recurrent modeling is highly effective for this dataset.
- SkipGram+GRU and SkipGram+BiGRU closely follow, showing that gated recurrent designs provide robust performance.
- TF–IDF baselines (TFIDF+DNN and TFIDF+LogReg) remain competitive but do not surpass the best Skip-gram sequence models in this run.
- The weakest model, SkipGram+SimpleRNN, exhibits very low Macro F1-score and accuracy, consistent with limited long-range dependency handling in vanilla RNNs.

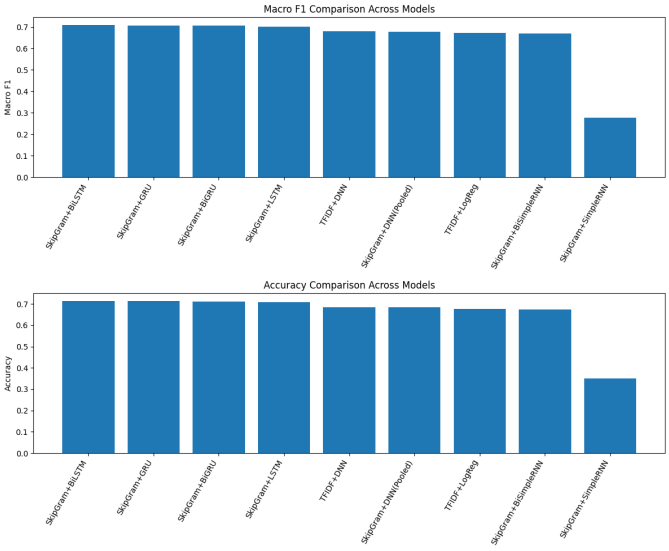


Fig. 3. Macro F1 Comparison Across Models (bar chart)

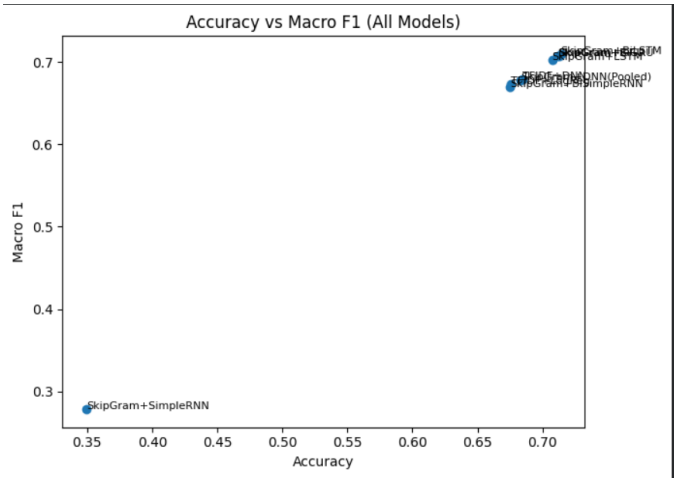


Fig. 4. Accuracy vs Macro F1 (scatter plot)

I. Best and Worst Models

This subsection highlights the highest- and lowest-performing models based on test-set evaluation metrics.

Best-performing model: SkipGram+BiLSTM

- Accuracy: 0.714012
- Macro F1-score: 0.709841

Worst-performing model: SkipGram+SimpleRNN

- Accuracy: 0.349172
- Macro F1-score: 0.278029

Among all evaluated models, SkipGram+BiLSTM achieves the highest performance on the test set, while SkipGram+SimpleRNN shows the weakest performance. The remaining models fall between these two extremes, indicating a clear performance gap between bidirectional recurrent architectures and basic recurrent models. **Interpretation:** The superior performance of the bidirectional LSTM model reflects

TABLE I
MODEL PERFORMANCE COMPARISON

Model	Accuracy	Macro-F1	Best Val Macro-F1	Best Config	Train Time (s)	Search Time
SkipGram+BiLSTM	0.714012	0.709841	0.712901	{max_len: 200, units: 64, drop: 0.5, lr: ...}	400.842938	2435.54730
SkipGram+GRU	0.712645	0.707617	0.713346	{max_len: 200, units: 64, drop: 0.5, lr: ...}	223.554771	1643.98275
SkipGram+BiGRU	0.711712	0.706961	0.711089	{max_len: 100, units: 32, drop: 0.5, lr: ...}	232.097140	1965.88059
SkipGram+LSTM	0.707378	0.702067	0.699107	{max_len: 100, units: 64, drop: 0.3, lr: ...}	166.577330	963.28325
TFIDF+DNN	0.684295	0.679488	0.684984	{hidden: [128], drop: 0.3, lr: 0.001}	34.899464	122.93689
SkipGram+DNN (Pooled)	0.683178	0.677654	0.680170	{max_len: 100, units: 64, drop: 0.3, lr: ...}	55.848236	480.26380
TFIDF+LogReg	0.675611	0.673430	0.677763	{C: 1.0}	15.823938	54.675704
SkipGram+BiSimpleRNN	0.674945	0.669367	0.674518	{max_len: 200, units: 64, drop: 0.3, lr: ...}	359.562831	2252.51171
SkipGram+SimpleRNN	0.349172	0.278029	0.279197	{max_len: 100, units: 64, drop: 0.5, lr: ...}	106.886188	1021.91280

its ability to utilize contextual information from both directions of the input sequence, which is particularly beneficial for question–answer text where relevant cues may appear at different positions. Conversely, the SimpleRNN model struggles to maintain useful information across longer sequences, leading to significantly lower classification performance.

J. Confusion Matrix and Classification Report

This subsection examines class-level prediction behavior using confusion matrices and classification reports. **Required visualizations:**

- Figure 5: Confusion Matrix for SkipGram+BiLSTM (Test Set)

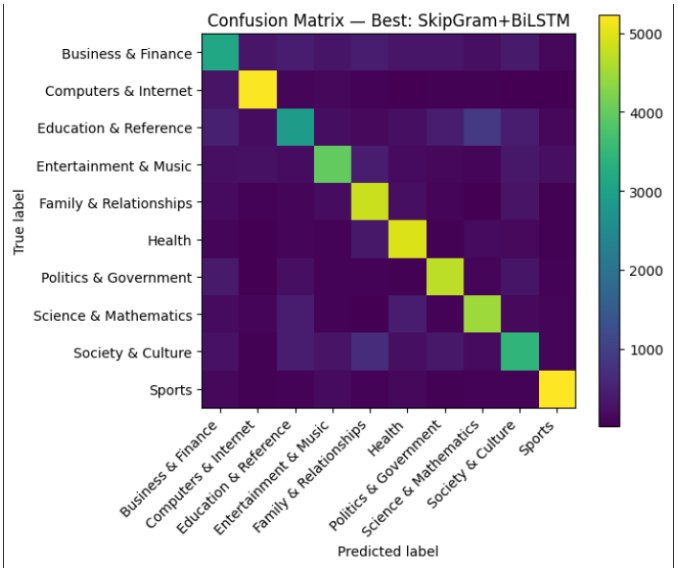


Fig. 5.

Classification Report for SkipGram+BiLSTM (placed here or in the Appendix)

Figure 6: Confusion Matrix for SkipGram+SimpleRNN (Test Set)

The confusion matrix for SkipGram+BiLSTM shows strong diagonal dominance, indicating that most samples are correctly classified across all categories. In contrast, the confusion matrix for SkipGram+SimpleRNN exhibits a weaker diagonal pattern with a higher number of off-diagonal entries, reflecting frequent misclassifications and less reliable class separation. The classification report complements these observations by presenting class-wise precision, recall, and F1-scores for the best-performing model, showing variation in performance across classes while maintaining stable overall results.

K. Best Machine Learning vs Best Neural Network Comparison

This subsection compares the performance of the best machine learning model with the best-performing neural network model. **Best Machine Learning Model (TF–IDF family): TFIDF+DNN**

- Accuracy: 0.684295
- Macro F1-score: 0.679488
- Best Validation Macro F1-score: 0.684984
- Best Configuration: {hidden: [128], dropout: 0.3, learning rate: 0.001}

Best Neural Network Model (Skip-gram family): Skip-Gram+BiLSTM

- Accuracy: 0.714012
- Macro F1-score: 0.709841
- Best Validation Macro F1-score: 0.712901
- Best Configuration includes max_len = 200, units = 64, dropout = 0.5

Required visualization:

- Figure 7: Best ML vs Best NN Comparison (Accuracy and Macro F1-score)

The comparison shows that the best neural network model achieves higher accuracy and Macro F1-score than the best machine learning model. While the machine learning model demonstrates competitive performance, it does not surpass the neural model on either evaluation metric.

L. Summary of Findings

- **Best Overall Model:** Skip-gram + Bidirectional LSTM
- **Best Neural Embedding Model:** Skip-gram + Bidirectional LSTM
- **Worst Performing Model:** Skip-gram + SimpleRNN

M. Key Findings

- Skip-gram–based neural models outperform TF–IDF–based approaches.
- Bidirectional recurrent models achieve the highest classification performance.

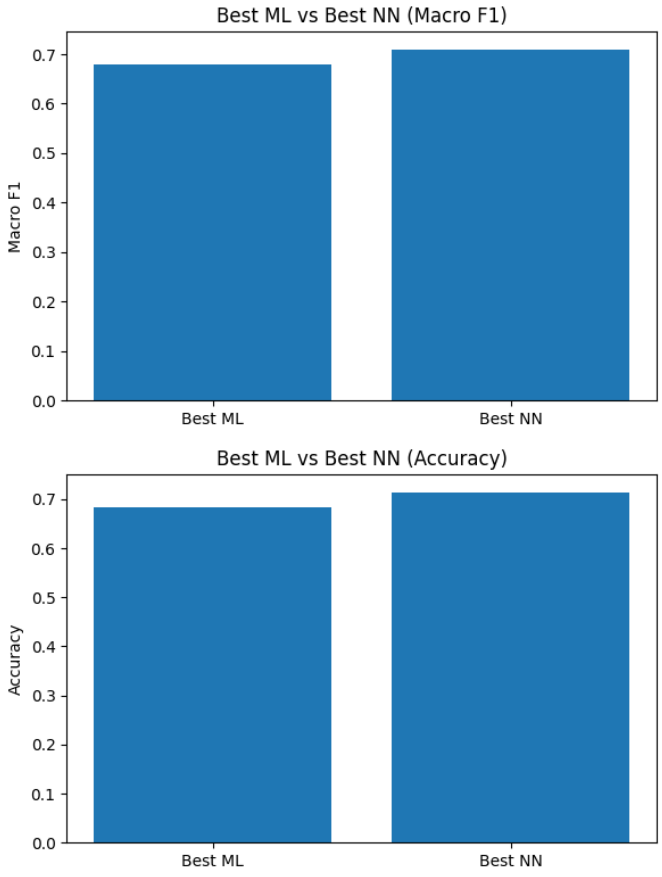


Fig. 6.

- SimpleRNN exhibits significantly weaker results than gated and bidirectional variants.
- The best model correctly predicts most classes, as reflected in the confusion matrix.
- Neural models outperform the machine learning baseline in both accuracy and Macro F1-score.

IV. Discussion

The experimental results provide clear evidence that both word representation and model architecture play a crucial role in multi-class question classification. Across all experiments, models using Skip-gram embeddings generally outperform those relying on TF–IDF representations, indicating that distributed semantic representations are more effective for capturing the underlying meaning of question–answer text. Unlike sparse lexical features, word embeddings encode semantic similarity between words, which helps models generalize better when different classes share overlapping vocabulary. Among the neural models, architectures designed to capture sequential information consistently perform better than feed-forward models. This trend highlights the importance of word order and contextual dependencies in question–answer data, where relevant cues may appear anywhere within the text. In particular, bidirectional recurrent models achieve the strongest performance, suggesting that access to both preceding and following context improves classification decisions. This is especially useful for longer QA texts, where early and late parts of the sequence may jointly determine the correct category. The poor performance of the SimpleRNN model further emphasizes the limitations of basic recurrent structures. Despite using the same Skip-gram embeddings as stronger models, SimpleRNN struggles to maintain meaningful information across longer sequences. This behavior is reflected in both its low overall scores and its confusion matrix, which shows widespread misclassification. In contrast, gated recurrent units such as GRU and LSTM provide more stable learning by controlling information flow, leading to noticeably improved results. The comparison between the best machine learning model and the best neural model reveals that traditional approaches remain competitive but are ultimately limited in this setting. TF–IDF-based models perform reasonably well, particularly for categories with distinctive keywords, but they fail to capture deeper contextual relationships. Neural sequence models, by contrast, are better suited to handling complex sentence structures and contextual cues, which explains their consistent performance Conclusion

This study presents a comparative evaluation of different word representation techniques and classification models for multi-class question–answer text classification. Through systematic experimentation, the results demonstrate that model performance is strongly influenced by both the choice of text representation and the underlying architecture. Neural models using Skip-gram embeddings consistently achieve higher performance than TF–IDF-based approaches, particularly when combined with recurrent and bidirectional structures.advantage. Analysis of confusion matrices and classification reports indicates that most misclassifications occur between semantically related categories, suggesting challenges in separating closely overlapping topics.

Finally, although embedding-based neural models achieve the best performance, training complexity and computational cost limit extensive hyperparameter exploration. Future work may benefit from richer contextual representations or more advanced architectures to further improve class discrimination.

V. Conclusion

This study presents a comparative evaluation of different word representation techniques and classification models for multi-class question–answer text classification. Through systematic experimentation, the results demonstrate that model performance is strongly influenced by both the choice of text representation and the underlying architecture. Neural models using Skip-gram embeddings consistently achieve higher performance than TF–IDF-based approaches, particularly when combined with recurrent and bidirectional structures. Bidirectional recurrent models achieve the most reliable performance, while simpler recurrent models show clear limitations. Traditional machine learning approaches remain competitive but are less effective at capturing contextual and sequential information in longer QA texts, and most classification errors occur between semantically related categories. Despite these

improvements, the study is subject to certain limitations, including computational constraints that restrict extensive hyperparameter tuning and the use of static word embeddings. Future work may explore contextualized embeddings or more advanced architectures to further enhance classification performance. Overall, the findings provide practical insight into how representation and architecture choices affect multi-class text classification outcomes.

REFERENCES